MILITARY SPECIFICATION SHEET

ELECTRON TUBES, CATHODE RAY

TYPES 3JP1, 3JP2, 3JP7, 3JP11, and 3JP12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Type	Ef	Ec1	ed	Eb1	Eb2	Eb3	Rg	Zd	Ehk	Eb3/Eb2	Alt
Unit:		V	Vdc	v	Vdc	Vdc	Vdc	Meg	Meg	Vdc	Ratio	ft
Maximum:	P1, P2, P11:	6.9	0	550	1,100	2,200	4,400	1.5	1.5	-125	2.3	50,000
	P7, P12:	6.9	0	550	1,100	2,200	4,400	1.5	1.0	-125	2.3	50,000
Minimum:	All:	5.7	-200	---	---	1,500	3,000	---	---	---	---	---

TEST CONDITIONS:

P1, P2, P11, P12:	6.3	Adj	---	Focus	1,500	3,000	---	---	---	---	---
P7:	6.3	Adj	---	Focus	2,000	4,000	---	---	---	---	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	SYMBOL	LIMITS MINIMUM	LIMITS MAXIMUM	UNIT
	Qualification inspection						
1002	Barometric pressure, reduced	All	87.0 mm Hg	---	---	---	---
5101	Neck and bulb alinement (electrostatic types)	All		dia	---	2.25	in.
5246	Focusing voltage, zero bias	P1, P2, P11		Eb1	300	515	Vdc
		P7		Eb1	400	590	Vdc
		P12		Eb1	302	517	Vdc
5248	Deflection factor (1102)	P1, P2 P11	Eb3 = Eb2 = 1,500 Vdc	DF	102	138	Vdc/in.
		P7	Eb3 = Eb2 = 2,000 Vdc	DF	136	184	Vdc/in.
		P12	Eb3 = Eb2 = 1,500 Vdc.	DF	96	144	Vdc/in.
5248	Deflection factor (3004)	P1,P2, P11	Eb3 = Eb2 = 1,500 Vdc	DF	76	102	Vdc/in.
		P7	Eb3 = Eb2 = 2,000 Vdc	DF	100	138	Vdc/in.
		P12	Eb3 = Eb2 = 1,500 Vdc	DF	71	107	Vdc/in.
5248	Deflection factor uniformity	All		---	---	---	---
5216	Cathode illumination	All		---	---	---	---
1331	Direct-interelectrode capacitance	All	g1 to all	Cg1	---	10.5	pf
			k to all	Ck	---	10.5	pf
			D1 to D2	C1D2	---	4.0	pf
			D3 to D4	C3D4	---	3.5	pf
			D1 to all	CD1	---	15.0	pf
			D2 to all	CD2	---	15.0	pf
			D3 to all	CD3	---	12.0	pf
			D4 to all	CD4	---	12.0	pf
1141	Pressure (implosion)	All		---	---	---	---
5111	Vibration	All		Width	---	1.0	mm
	Quality conformance inspection, part 1						
5201	Voltage breakdown	All		---	---	---	---
5201	Voltage breakdown (electrostatic types)	All		---	---	---	---
5206	Gas "cross"	P1,P2	Ib3 = 100 µAdc	---	---	---	---
		P11	(see note 7)				
		P7	Ib3 = 200 µAdc (see note 7)	---	---	---	---
		P12	Ib3 = 30 µAdc (see note 7)	---	---	---	---
5106	Bulb, screen, and faceplate quality	All		---	---	---	---
---	Deflection plate cutoff	P1,P2 P7,P11	Ib3 = 10 µAdc (max) (see note 3)	---	---	---	---
		P12	Ib3 = 5 µAdc (max) (see notes 3 and 4)	---	---	---	---
5223	Modulation	P7	Ib3 = 200 µAdc	ΔEc	---	55	Vdc
		P12	Ib3 = 100 µAdc	ΔEc	---	40	Vdc

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1 -Continued						
5231	Spot position (electrostatic deflection)	P1,P2 P11,P12 P7		--- ---	--- ---	15 12	mm mm
5231	Spot displacement (leakage)	All		Displ	---	7	mm
5241	Grid-cutoff voltage	P1,P2 P11,P12 P7		Ec1 Ec1	-22.5 -30	-67.5 -90	Vdc Vdc
5251	Grid No. 1 leakage current	All		---	---	---	---
5251	Anode No. 2 leakage current	All		---	---	---	---
5221	Light output	P1 P11	Eb2 = 1,500 V; Eb3 = 3,000 V; Ib3 = 100 µAdc Eb2 = 1,500 V; Eb3 = 3,000 V; Ib3 = 100 µAdc (see note 5)	Light Light	25 9	--- ---	ftL ftL
	Quality conformance inspection, part 2						
1301	Heater current	All		If	540	660	mA
5201	Electrode current (anode No. 1)	P1,P2 P11 P7 P12	Ib3 = 100 µAdc Ib3 = 200 µAdc Ib3 = 5 µAdc (see note 1)	Ib1 Ib1 Ib1	-50 -50 -50	10 10 10	µAdc µAdc µAdc
5101	Base alinement (electrostatic types)	All	+102, pin No. 5	---	---	---	---
5101	Side terminal alinement (electrostatic types)	All	+102	---	---	---	---
5201	Electrode current (anode No. 3)	P1,P2, P11	See note 2	Ib3	150	---	µAdc
5201	Electrode current (cathode)	P1,P2, P11 P7 P12	Ib3 = 100 µAdc Ib3 = 200 µAdc Ib3 = 5 µAdc (see note 1)	Ik Ik Ik	--- --- ---	1,000 1,500 100	µAdc µAdc µAdc
5101	Angle between traces	All		---	---	---	---
5101	Neck and base alinement (electrostatic types)	All		---	---	---	---
5216	Stray light emission (conventional types)	All	Eb2 = 2,200 Vdc; Eb3 = 4,400 Vdc	---	---	---	---
5221	Screens	P2 P7 P12		D1(1) D1(5) GS:1 cbf D0.1(E)- D0.3(E)	360 400 4 550 80	--- --- --- --- 100	cB cB --- cB

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METHOD	REQUIREMENT OR TEST	TYPE	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued						
5226	Line width "A" (electrostatic deflection)	P1,P2, P11	Ib3 = 100 µAdc (see note 6)	Width	---	0.75	mm
		P7	Ib3 = 200 µAdc (see note 6)	Width	---	0.80	mm
		P12	Ib3 = 5 µAdc (see notes 1 and 6)	Width	---	0.35	mm
5226	Line width "B" (electrostatic deflection)	P1,P2, P11	Ib3 = 100 µAdc (see note 8)	Width	---	0.90	mm
		P7	Ib3 = 200 µAdc (see note 8)	Width	---	1.00	mm
		P12	Ib3 = 5 µAdc (see notes 1 and 8)	Width	---	0.40	mm
5230	Focusing voltage at cutoff	P1,P2, P11		Eb1	345	515	Vdc
		P7		Eb1	460	690	Vdc
		P12		Eb1	345	517	Vdc
5248	Deflection factor (102)	P1,P2, P11,P12		DF	127	173	Vdc/in.
		P7		DF	170	230	Vdc/in.
5248	Deflection factor (304)	P1,P2, P11,P12		DF	94	128	Vdc/in.
		P7		DF	125	170	Vdc/in.
5251	Heater-cathode leakage current	All		---	---	---	---
1101	Secureness of base, cap, or insert	All		---	---	---	---
1105	Permanence of marking			---	---	---	---
1111	Base pin solder depth			---	---	---	---
	Quality conformance inspection, part 3						
---	Life-test provisions	P1,P2 P7,P11	Group C; t = 500 hours (min) Eb3 = 4,400 Vdc; Eb2 = 2,200 Vdc; Ib3 = 30 µAdc	---	---	---	---
		P12	Group C; t = 500 hours (min) Eb3 = 4,400 Vdc; Eb2 = 2,200 Vdc; Ib3 = 5 µAdc	---	---	---	---
---	Life test end points:						
5223	Modulation	P1,P2, P11,P12	Ib3 = 100 µAdc;	ΔEc	---	40	Vdc
		P7	Ib3 = 150 µAdc;	ΔEc	---	55	Vdc
5226	Line width "A" (electrostatic deflection)	P1,P2, P11	Ib3 = 100 µAdc;	Width	---	0.75	mm
		P7	Ib3 = 150 µAdc;	Width	---	0.86	mm
		P12	Ib3 = 5 µAdc;	Width	---	0.35	mm
5226	Line width "B" (electrostatic deflection)	P1,P2, P11	Ib3 = 100 µAdc;	Width	---	0.90	mm
		P7	Ib3 = 150 µAdc;	Width	---	1.12	mm
		P12	Ib3 = 5 µAdc;	Width	---	0.40	mm
1331	Direct-interelectrode capacitance	All	g1 to all	Cg1	---	10.5	pF
			k to all	Ck	---	10.5	pF
			D1 to D2	C1D2	---	4.0	pF
			D3 to D4	C3D4	---	3.5	pF
			D1 to all	CD1	---	15.0	pF
			D2 to all	CD2	---	15.0	pF
			D3 to all	CD3	---	12.0	pF
			D4 to all (See note 10)	CD4	---	12.0	pF

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NOTES:

1. The tube can be severely and permanently damaged if the current density on the screen is allowed to rise too high in static tests of this sort. For this reason, the length of time during which the screen is bombarded should be kept as short as possible compatible with the reading of specified currents limits.

2. Ib3 is to be measured with ΔEc=40 Vdc when cutoff is greater than -40 Vdc and with Ec1=0 when cutoff is less than -40 Vdc.

3. At specified conditions there shall be no pattern cutoff with the useful screen area.

4. To prevent burning of the screen, the beam current should not be allowed to exceed that amount necessary to just produce a visible trace.

5. As measured by a 2x2 inch (50.80 x 50.80 mm) raster using a type 3 photronic cell without eye correction, calibrated in foot candles of illumination from a light source having color temperature of 2,700°K.

6. The deflecting plates shall be returned to anode No. 2 through a 2.5 megohm resistance. Ib3, the beam current, shall be set as follows:

P1 - - - - - -	Ib3 = 100 μAdc	P11 - - - - - -	Ib3 = 100 μAdc
P2 - - - - - -	Ib3 = 100 μAdc	P12 - - - - - -	Ib3 = 5 μAdc
P7 - - - - - -	Ib3 = 200 μAdc		

   a. The high-frequency scanning shall be applied to the deflecting plates nearest the screen and the amplitude shall be adjusted to give a line length of approximately 90 percent of the maximum tube diameter. The low-frequency scanning amplitude shall be expanded to approximately 90 percent of the maximum tube diameter in the direction perpendicular to the direction of high-frequency scanning. Readjustment may be made for best overall focus. The tubes shall be observed for deflection defocusing, astigmatism, or spot ellipticity observable to the eye as evidenced by fuzziness due to lack of sharpness of the trace (usually around the edges), bow-tieing (irregular widths of any single line when observed at different points), bowing of trace other than that normally caused by curvature of bulb.

   b. This test for focus is to be made in addition to the line width measurements.

7. This test to be performed at the conclusion of the holding period.

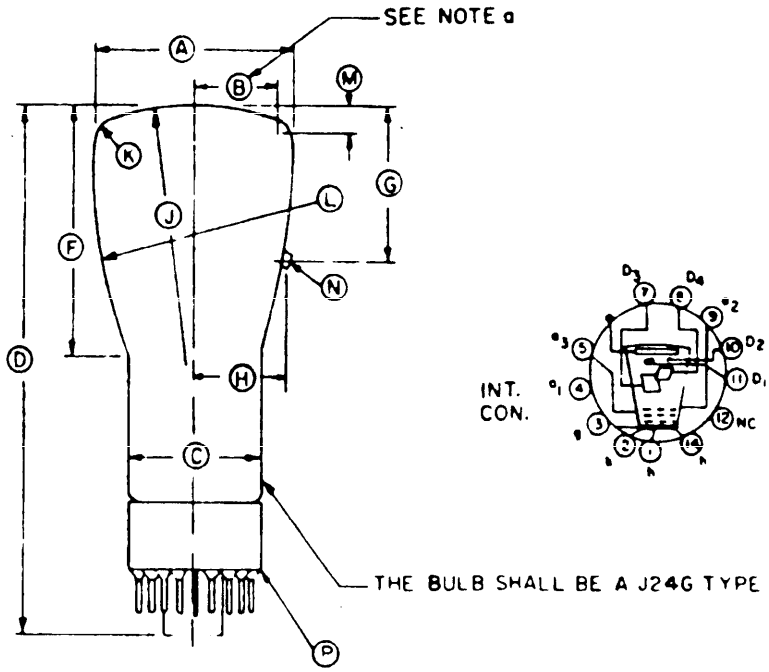
8. The same conditions shall be set up as described in note 6 except that the connection of deflection elements to the low- and high-frequency scanning supplies shall be interchanged and the amplitudes adjusted to 90 percent of the maximum tube diameter in both directions without any adjustment of focus from note 6 conditions.

   An examination for defocusing, astigmatism, or spot ellipticity shall be made as in note 6.

9. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

10. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses (see note b)	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	2.94 (74.7)	3.06 (77.7)
B	1.37 (34.8)	
C	1.94 (49.3)	2.06 (52.3)
D	9.75 (247.6)	10.25 (260.4)
F	3.75 (95.2)	4.00 (101.6)
G	1.50 (38.1)	2.00 (50.8)
Quality conformance inspection, part 3 (See note 10)		
N	Bulb contact: J1-22	
P	Base: B12-37	
Reference dimensions (see note c)		
H	1.62 (41.1)	
J	8.00 (203.2) R	
K	.25 (6.4) R	
L	12.44 (316.0) R	
M	.35 (8.9)	

NOTES:
a. The minimum useful screen radius shall not be less than 1.375 inches (34.92 mm).
b. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
c. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube types 3JP1, 3JP2, 3JP7, 3JP11, and 3JP12.

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Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activities:
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Preparing activity:
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